1 Lie Algebras and Representation Theory

Basic

Exercise 1.1. *Let W_1 be the Witt algebra spanned by the basis $\{d_n = x^{n+1}\frac{\mathrm{d}}{\mathrm{d}x} \mid n \in \mathbb{Z}\}$ with the Lie bracket relations:*

$$[d_n, d_m] = (m-n)d_{n+m}$$

1. *Consider the subspace $\mathfrak{h} = \mathrm{span}_{\mathbb{C}}\{d_{-1}, d_0, d_1\}$. By explicitly calculating the brackets $[d_0, d_{-1}]$, $[d_0, d_1]$, and $[d_1, d_{-1}]$, prove that $\mathfrak{h}$ is closed under the Lie bracket (i.e., it is a Lie subalgebra).*

2. *The standard basis $\{e, f, h\}$ of $\mathfrak{sl}(2, \mathbb{C})$ satisfies the relations $[h, e] = 2e$, $[h, f] = -2f$, and $[e, f] = h$. Give an explicit isomorphism of Lie algebras from $\mathfrak{sl}(2, \mathbb{C})$ to $\mathfrak{h}$.*

Exercise 1.2. *Let $A_1 = \langle x, y \mid yx - xy = 1\rangle$ be the first Weyl algebra (over $\mathbb{C}$), $d = yx$.*

1. *Using the algebraic relations, rewrite x^2y^2 as a polynomial in d. That is, find constants a, b, c such that:*

$$x^2y^2 = ad^2 + bd + c$$

2. *Show that $\{x^iy^j | i, j \in \mathbb{N}\}$ is a basis of A_1 as a vector space over $\mathbb{C}$.*

Intermediate

Exercise 1.3. *Recall the definition of the modules $V(\alpha, \beta)$ for the Witt algebra W_1. The underlying vector space is $z^{\beta}\mathbb{C}[z, z^{-1}](\mathrm{d}z)^{\alpha}$ with the action:*

$$d_n \cdot (z^{\gamma}(\mathrm{d}z)^{\alpha}) = (\gamma + n\alpha)z^{\gamma+n}(\mathrm{d}z)^{\alpha}$$

1. *Check whether $V(-1, 0)$ is irreducible. If it is, prove it. If it is not, explicitly identify a non-trivial proper submodule.*

2. *Show that $V(1, 0)$ is reducible, describe its maximal proper submodule, denoted by W.*

3. *Prove that $W \simeq V(0, 0)/\mathbb{C} \cdot 1$.*

Exercise 1.4. *According to the Clebsch-Gordan decomposition theory, which is for simple $\mathfrak{sl}(2, \mathbb{C})$ modules $V(n)$ and $V(m)$ where $n \geq m$, $V(n) \otimes V(m) \simeq V(n+m) \oplus V(n+m-2) \oplus \ldots \oplus V(n-m)$.*

1. *Decompose the tensor product of the irreducible* $\mathfrak{sl}(2,\mathbb{C})$*-modules* $V(3)\otimes V(1)$ *into a direct sum of irreducible modules.*

2. *Let* v_0 *be the highest weight vector of* $V(3)$ *(weight 3) and* u_0 *be the highest weight vector of* $V(1)$ *(weight 1). In the decomposition above, the highest weight component (isomorphic to* $V(4)$*) is generated by* $v_0 \otimes u_0$*.*

   Using the action of the lowering operator f *(via the rule* $f(v\otimes u) = fv\otimes u + v\otimes fu$*), find the specific linear combination of basis vectors that represents the* ***highest weight vector*** *of the component isomorphic to* $V(2)$ *(i.e., a vector of weight 2 that is annihilated by* e*).*

Advanced

Exercise 1.5. *Recall that the Casimir element in the universal enveloping algebra* $U(\mathfrak{sl}(2,\mathbb{C}))$ *is defined as* $c = (h+1)^2 + 4fe$*. It is known that* c *acts as a scalar* $(\lambda+1)^2$ *on any irreducible module with highest weight* λ*.*

Consider the tensor product module $W = V(1)\otimes V(1)\otimes V(1)$ *(note that the action of* $x \in \mathfrak{sl}(2,\mathbb{C})$ *on triple tensor is* $x\cdot(v\otimes u\otimes w) = xv\otimes u\otimes w + v\otimes xu\otimes w + v\otimes u\otimes xw$*). Let* $\{v_1, v_{-1}\}$ *be the standard weight basis for the first* $V(1)$*,* $\{u_1, u_{-1}\}$ *for the second* $V(1)$*,* $\{w_1, w_{-1}\}$ *for the third* $V(1)$ *(indices denote weights). The space* W *has a basis* $\{v_1\otimes u_1\otimes w_1, v_1\otimes u_1\otimes w_{-1}, v_1\otimes u_{-1}\otimes w_1, v_1\otimes u_{-1}\otimes w_{-1}, v_{-1}\otimes u_1\otimes w_1, v_{-1}\otimes u_1\otimes w_{-1}, v_{-1}\otimes u_{-1}\otimes w_1, v_{-1}\otimes u_{-1}\otimes w_{-1}\}$.

1. *Write down the decomposition of* W *into a direct sum of irreducible submodules.*

2. *How many highest weight vectors are there in* W*? Write down explicit expressions for each of them in terms of the product basis above.*

3. *Using the results above, determine the matrix representation of the operator* c *acting on* W *with respect to the product basis.*

2 Fnite Group Representation Theory

Basic

Exercise 2.1. *Let $G = S_3$ and let V be the standard 2-dimensional irreducible representation. The character values for V on the three conjugacy classes e, (12), and (123) are given by:*

$$\chi_V(e) = 2, \quad \chi_V((12)) = 0, \quad \chi_V((123)) = -1$$

Consider the tensor product representation $\rho = V \otimes V$.

(a) *Compute the character χ_ρ for all conjugacy classes and calculate the inner product $\langle \chi_\rho, \chi_\rho \rangle$.*

(b) *Using the orthogonality relations, calculate the multiplicity of the trivial representation* $\mathbf{1}$ *in the decomposition of ρ. (Recall that $\chi_{\mathbf{1}}(g) = 1$ for all g).*

Intermediate

Exercise 2.2. *Let $G = S_3$. Consider the set $X = \{(i,j) \mid 1 \leq i,j \leq 3, i \neq j\}$ of ordered pairs of distinct elements from $\{1,2,3\}$. The size of X is 6. S_3 acts on X naturally by $\sigma \cdot (i,j) = (\sigma(i), \sigma(j))$. Let π be the associated permutation representation on the vector space $\mathbb{C}[X]$.*

(a) *Determine the character χ_π of this representation by counting the fixed points for representatives of each conjugacy class: e, (12), and (123).*

(b) *Decompose χ_π into a sum of irreducible characters of S_3.*

Advanced

Exercise 2.3. *Let $G = S_4$ and let W be the standard 3-dimensional irreducible representation defined by:*

$$W = \left\{ (x_1, x_2, x_3, x_4) \in \mathbb{C}^4 \mid \sum_{i=1}^{4} x_i = 0 \right\}$$

where G acts by permuting the coordinates.

Let $H = S_3$ be the subgroup of G consisting of permutations that fix the index 4 (i.e., permutations of $\{1,2,3\}$). Consider the permutation representation ℓ^H on the coset space $\mathbb{C}[G/H]$.

(a) *The multiplicity of an irreducible representation V_k in ℓ^H is given by $m_k = \dim(V_k^{*H})$, where V_k^{*H} is the subspace of vectors in V_k^* fixed by all elements of H.*

Explicitly construct the subspace $W^H = \{w \in W \mid \sigma \cdot w = w, \forall \sigma \in H\}$.

(b) *Based on your construction in part (a), determine the dimension $\dim(W^H)$ and thus the multiplicity of the standard representation W in the decomposition of ℓ^H.*

3 Solutions

3.1 Lie Algebra and Representation Theory

Solution to Problem 1

1. Proof that $\mathfrak{h}$ is a Lie subalgebra:

We calculate the Lie brackets for the basis elements d_{-1}, d_0, d_1 using the defining relation $[d_n, d_m] = (m-n)d_{n+m}$.

$$\begin{aligned}
[d_0, d_{-1}] &= (-1-0)d_{0+(-1)} = -d_{-1} \in \mathfrak{h} \\
[d_0, d_1] &= (1-0)d_{0+1} = d_1 \in \mathfrak{h} \\
[d_1, d_{-1}] &= (-1-1)d_{1+(-1)} = -2d_0 \in \mathfrak{h}
\end{aligned}$$

Since the result of the bracket operation between any two basis vectors lies within the span of the basis vectors $\{d_{-1}, d_0, d_1\}$, the subspace $\mathfrak{h}$ is closed under the Lie bracket. Thus, $\mathfrak{h}$ is a Lie subalgebra of W_1.

2. Explicit Isomorphism:

Let $\phi : \mathfrak{sl}(2, \mathbb{C}) \to \mathfrak{h}$ be a linear map defined on the standard basis $\{e, f, h\}$ as follows:

$$\phi(e) = d_1, \quad \phi(f) = -d_{-1}, \quad \phi(h) = 2d_0.$$

We verify that this map preserves the Lie bracket relations of $\mathfrak{sl}(2, \mathbb{C})$:

- **Relation $[h, e] = 2e$:**

$$[\phi(h), \phi(e)] = [2d_0, d_1] = 2[d_0, d_1] = 2(d_1) = 2\phi(e). \quad \checkmark$$

- **Relation $[h, f] = -2f$:**

$$[\phi(h), \phi(f)] = [2d_0, -d_{-1}] = -2[d_0, d_{-1}] = -2(-d_{-1}) = 2d_{-1}.$$

  On the other hand, $-2\phi(f) = -2(-d_{-1}) = 2d_{-1}$. Thus, the relation holds. ✓

- **Relation $[e, f] = h$:**

$$[\phi(e), \phi(f)] = [d_1, -d_{-1}] = -[d_1, d_{-1}] = -(-2d_0) = 2d_0 = \phi(h). \quad ✓$$

Thus, the map ϕ is an isomorphism of Lie algebras.

Solution to Problem 2

1. Rewriting x^2y^2 in terms of d:

We use the relation $xy = yx - 1 = d - 1$. First, we establish the commutation relation between x and d. Since $d = yx$:

$$dx = (yx)x = yx^2 = (xy + 1)x = x(yx) + x = xd + x = x(d + 1).$$

(Alternatively, using the commutator $[d, x] = y[x, x] + [y, x]x = 0 + 1 \cdot x = x$, so $dx - xd = x$).

Now we compute x^2y^2:

$$\begin{aligned}
x^2y^2 &= x(xy)y \\
&= x(d - 1)y \\
&= (xd)y - xy \\
&= (dx - x)y - (d - 1) \\
&= d(xy) - xy - d + 1 \\
&= d(d - 1) - (d - 1) - d + 1 \\
&= d^2 - d - d + 1 - d + 1 \\
&= d^2 - 3d + 2.
\end{aligned}$$

Thus, the constants are $a = 1, b = -3, c = 2$.

2. Basis of A_1:

A_1 is defined as the quotient of the free algebra $\mathbb{C}\langle x, y\rangle$ by the ideal $I = \langle yx - xy - 1\rangle$. We assume a filtration on A_1 where $\deg(x) = \deg(y) = 1$. The associated graded algebra $\mathrm{gr}(A_1)$ is generated by $\bar{x}, \bar{y}$ with the relation:

$$\bar{y}\bar{x} - \bar{x}\bar{y} = 0$$

(since the non-commutative term 1 has degree 0, which is strictly lower than degree 2). Thus, $\mathrm{gr}(A_1) \cong \mathbb{C}[x, y]$ (the commutative polynomial ring).

The set of monomials $\{x^i y^j \mid i, j \in \mathbb{N}\}$ forms a basis for the commutative polynomial ring $\mathbb{C}[x, y]$. By the standard argument for filtered algebras (analogous to the Poincaré-Birkhoff-Witt theorem), the set of ordered monomials $\{x^i y^j \mid i, j \in \mathbb{N}\}$ forms a vector space basis for A_1.

Solution to Problem 3

1. Irreducibility of $V(-1, 0)$:

The action is given by $d_n \cdot v_k = (k - n)v_{k+n}$ (where we identify $v_k = z^k (\mathrm{d}z)^{-1}$). This module is isomorphic to the adjoint representation of W_1 (via $v_k \leftrightarrow d_k$). Since W_1 is simple, its adjoint representation is irreducible.

Direct Proof: Let U be a non-zero submodule. Let $v = \sum a_k v_k \in U$ ($v \neq 0$). The operator d_0 acts diagonally ($d_0 v_k = k v_k$), so we can isolate individual basis vectors $v_k \in U$.

- Apply d_{-k} to v_k: $d_{-k} v_k = (k - (-k))v_0 = 2k v_0$. If $k \neq 0$, we generate v_0.
- If we start with v_0, applying d_n gives $d_n v_0 = (0 - n)v_n = -n v_n$. This generates all v_n for $n \neq 0$.
- If we start with v_{-1}, apply d_1: $d_1 v_{-1} = (-1 - 1)v_0 = -2v_0$.

Thus, from any non-zero vector, we can generate the entire space. $V(-1, 0)$ is irreducible.

2. Reducibility of $V(1, 0)$:

The action is $d_n \cdot z^k \mathrm{d}z = (k + n)z^{k+n} \mathrm{d}z$. Let $v_k = z^k \mathrm{d}z$. Consider the vector $v_0 = \mathrm{d}z$ (index $k = 0$).

$$d_n \cdot v_0 = (0 + n)v_n = n v_n.$$

v_0 generates the whole module (for $n \neq 0$). However, can we reach v_0 from other vectors? Suppose we have v_k with $k \neq 0$. Applying d_{-k}:

$$d_{-k} \cdot v_k = (k + (-k))v_0 = 0 \cdot v_0 = 0.$$

The coefficient is always 0 when the target index is 0. Thus, v_0 cannot be generated by any other basis vector. The subspace spanned by all vectors *except* v_0 is invariant. Let $W = \mathrm{span}_{\mathbb{C}}\{z^k \mathrm{d}z \mid k \in \mathbb{Z}, k \neq 0\}$.

- If $k+n \neq 0$, $d_n v_k = (k+n)v_{k+n} \in W$.
- If $k+n = 0$, $d_n v_k = 0 \in W$.

Thus, W is a proper submodule, and $V(1,0)$ is reducible.

3. Isomorphism of W:

Consider the module $V(0,0) = \mathbb{C}[z, z^{-1}]$ with action $d_n z^k = kz^{k+n}$. The subspace $\mathbb{C} \cdot 1$ ($k = 0$) is a trivial submodule. The quotient $M = V(0,0)/\mathbb{C}$ has a basis $\{[z^k] \mid k \neq 0\}$. The action on the quotient is $d_n[z^k] = k[z^{k+n}]$.

Define $\phi : W \to M$ by $\phi(z^k \mathrm{d}z) = \frac{1}{k}[z^k]$ for $k \neq 0$. We check equivariance:

$$\phi(d_n \cdot z^k \mathrm{d}z) = \phi((k+n)z^{k+n}\mathrm{d}z) = (k+n)\frac{1}{k+n}[z^{k+n}] = [z^{k+n}].$$

$$d_n \cdot \phi(z^k \mathrm{d}z) = d_n\left(\frac{1}{k}[z^k]\right) = \frac{1}{k}(k[z^{k+n}]) = [z^{k+n}].$$

The actions match, and ϕ is a bijection on the basis. Thus $W \simeq V(0,0)/\mathbb{C}$.

Solution to Problem 4

1. Decomposition:

Using the Clebsch-Gordan rule for $\mathfrak{sl}(2,\mathbb{C})$:

$$V(3) \otimes V(1) \cong V(3+1) \oplus V(3-1) = V(4) \oplus V(2).$$

2. Highest Weight Vector of $V(2)$:

Let v_k denote a basis vector of weight $3-2k$ in $V(3)$ (where v_0 is HW) and u_k denote a basis vector of weight $1-2k$ in $V(1)$ (where u_0 is HW). The action of the lowering operator f (unnormalized) is $fv_k \propto v_{k+1}$. Specifically, for the highest weight vectors: $fv_0 = C_1 v_1$ and $fu_0 = C_2 u_1$. For simplicity, we assume standard normalization where coefficients are non-zero.

The highest weight vector of the $V(4)$ component is $w_{V(4)} = v_0 \otimes u_0$. The vector in $V(4)$

with weight 2 (one step down) is:

$$f(v_0 \otimes u_0) = (fv_0) \otimes u_0 + v_0 \otimes (fu_0) = 3v_1 \otimes u_0 + v_0 \otimes u_1.$$

(Note: Using the convention $f(x^k) = (m-k)x^{k+1}$ for $V(m)$, we get coeff 3 for $V(3)$ and 1 for $V(1)$).

The highest weight vector ψ of the $V(2)$ component must lie in the same weight space (weight $3+1-2=2$) and be orthogonal to $f(v_0 \otimes u_0)$ (or equivalently, annihilated by e). Let $\psi = A(v_1 \otimes u_0) + B(v_0 \otimes u_1)$. Applying the raising operator e (with $ev_1 = v_0$ and $eu_1 = u_0$, up to scaling):

$$\begin{aligned} e\psi &= A(ev_1 \otimes u_0 + v_1 \otimes eu_0) + B(ev_0 \otimes u_1 + v_0 \otimes eu_1) \\ &= A(v_0 \otimes u_0) + B(v_0 \otimes u_0) \\ &= (A+B)(v_0 \otimes u_0). \end{aligned}$$

For ψ to be a highest weight vector, we need $e\psi = 0$, which implies $A = -B$. Thus, the vector is (up to a scalar):

$$\psi = v_1 \otimes u_0 - v_0 \otimes u_1.$$

Solution to Problem 5

1. Decomposition of $W = V(1) \otimes V(1) \otimes V(1)$:

First step: $V(1) \otimes V(1) \cong V(2) \oplus V(0)$. Second step:

$$\begin{aligned} (V(2) \oplus V(0)) \otimes V(1) &\cong (V(2) \otimes V(1)) \oplus (V(0) \otimes V(1)) \\ &\cong (V(3) \oplus V(1)) \oplus V(1). \end{aligned}$$

Total decomposition:

$$W \cong V(3) \oplus V(1) \oplus V(1).$$

2. Highest Weight Vectors:

There are 3 highest weight vectors. Let e_1 (weight 1) and e_{-1} (weight -1) be the basis of $V(1)$. Notation: $e_{ijk} = e_i \otimes e_j \otimes e_k$.

- **For $V(3)$:** The unique state with max weight $1+1+1=3$.

$$w_{V(3)} = e_{111}$$

- **For the two $V(1)$ components:** These must have weight 1 and be annihilated by e.

  First HW vector (via $V(2) \otimes V(1)$): The weight 0 state in $V(2)$ is proportional to $e_{1,-1} + e_{-1,1}$. Coupling $V(2)$ (weight 0) and $V(1)$ (weight 1) to get $V(1)$ requires the combination $v_{\text{mid}} \otimes u_{\text{high}} - v_{\text{high}} \otimes u_{\text{low}}$. Let $v_{\text{high}} = e_{11}$ and $v_{\text{mid}} = e_{1,-1} + e_{-1,1}$.

$$\psi_1 = (e_{1,-1} + e_{-1,1}) \otimes e_1 - 2(e_{11}) \otimes e_{-1} = e_{1,-1,1} + e_{-1,1,1} - 2e_{1,1,-1}.$$

  Second HW vector (via $V(0) \otimes V(1)$): The $V(0)$ singlet state is $e_{1,-1} - e_{-1,1}$. Coupling this with $V(1)$ is trivial (tensor product).

$$\psi_2 = (e_{1,-1} - e_{-1,1}) \otimes e_1 = e_{1,-1,1} - e_{-1,1,1}.$$

3. Matrix of the Casimir Operator c:
The Casimir element acts as a scalar $(\lambda+1)^2$ on $V(\lambda)$.

- On $V(3)$: eigenvalue $(3+1)^2 = 16$.
- On $V(1)$: eigenvalue $(1+1)^2 = 4$.

The matrix is block diagonal with respect to the weight spaces. We focus on the **weight 1 subspace**, spanned by $\{e_{1,1,-1}, e_{1,-1,1}, e_{-1,1,1}\}$. Let's denote these basis vectors as x, y, z respectively.

We know the spectral decomposition of the operator c on this subspace. One eigenvector is $v_3 = x+y+z$ (part of $V(3)$) with eigenvalue 16. Two eigenvectors (orthogonal complement) are in $V(1)$ with eigenvalue 4. The operator can be written as:

$$M = 16P_{V(3)} + 4P_{V(1)\oplus V(1)}.$$

Since $P_{V(1)\oplus V(1)} = I - P_{V(3)}$, we have $M = 4I + 12P_{V(3)}$. The projection onto the vector

$\mathbf{1} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$ is $P_{V(3)} = \frac{1}{3}\mathbf{1}\mathbf{1}^T = \frac{1}{3}\begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix}$. Thus,

$$M = \begin{pmatrix} 4 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 4 \end{pmatrix} + \begin{pmatrix} 4 & 4 & 4 \\ 4 & 4 & 4 \\ 4 & 4 & 4 \end{pmatrix} = \begin{pmatrix} 8 & 4 & 4 \\ 4 & 8 & 4 \\ 4 & 4 & 8 \end{pmatrix}.$$

The full matrix for W is block diagonal.

- On subspace $\{e_{111}\}$ (Weight 3): [16].
- On subspace $\{e_{1,1,-1}, e_{1,-1,1}, e_{-1,1,1}\}$ (Weight 1): $\begin{pmatrix} 8 & 4 & 4 \\ 4 & 8 & 4 \\ 4 & 4 & 8 \end{pmatrix}$.
- On subspace $\{e_{-1,-1,1}, e_{-1,1,-1}, e_{1,-1,-1}\}$ (Weight -1): $\begin{pmatrix} 8 & 4 & 4 \\ 4 & 8 & 4 \\ 4 & 4 & 8 \end{pmatrix}$.
- On subspace $\{e_{-1,-1,-1}\}$ (Weight -3): [16].

3.2 Finite Group Representation Theory

Solution to Problem 1

(a) Computing χ_ρ

According to Proposition 12.20, the character of a tensor product is the product of the characters:

$$\chi_\rho(g) = \chi_{V\otimes V}(g) = \chi_V(g) \cdot \chi_V(g) = (\chi_V(g))^2$$

We compute the values for each conjugacy class:

- Class e: $\chi_\rho(e) = (2)^2 = 4$.
- Class (12): $\chi_\rho((12)) = (0)^2 = 0$.
- Class (123): $\chi_\rho((123)) = (-1)^2 = 1$.

The sizes of the conjugacy classes in S_3 are: $|C_e| = 1$, $|C_{(12)}| = 3$, $|C_{(123)}| = 2$. The order of the group is $|G| = 6$.

Now, we calculate the inner product $\langle \chi_\rho, \chi_\rho \rangle$:

$$\begin{aligned}
\langle \chi_\rho, \chi_\rho \rangle &= \frac{1}{|G|} \sum_{g \in G} |\chi_\rho(g)|^2 \\
&= \frac{1}{6} \left(1 \cdot (4)^2 + 3 \cdot (0)^2 + 2 \cdot (1)^2\right) \\
&= \frac{1}{6}(16 + 0 + 2) \\
&= \frac{18}{6} = 3.
\end{aligned}$$

(b) Multiplicity of the Trivial Representation

Let $n_{\mathbf{1}}$ be the multiplicity of the trivial representation $\mathbf{1}$ in ρ. By Corollary 14.14:

$$n_{\mathbf{1}} = \langle \chi_\rho, \chi_{\mathbf{1}} \rangle$$

Recall that $\chi_{\mathbf{1}}(g) = 1$ for all g.

$$\begin{aligned}
\langle \chi_\rho, \chi_{\mathbf{1}} \rangle &= \frac{1}{6} \sum_{\text{classes}} |C_g| \cdot \chi_\rho(g) \cdot \overline{\chi_{\mathbf{1}}(g)} \\
&= \frac{1}{6} \left(1 \cdot 4 \cdot 1 + 3 \cdot 0 \cdot 1 + 2 \cdot 1 \cdot 1\right) \\
&= \frac{1}{6}(4 + 0 + 2) \\
&= \frac{6}{6} = 1.
\end{aligned}$$

Thus, the multiplicity of the trivial representation is **1**.

Solution to problem 2

(a) Determining the Character χ_π

For a permutation representation, the character value $\chi_\pi(g)$ is equal to the number of elements in the set X fixed by g.

$$\chi_\pi(g) = |\{x \in X \mid g \cdot x = x\}|$$

An element $(i, j) \in X$ is fixed by σ if and only if $\sigma(i) = i$ AND $\sigma(j) = j$.

- **Class e:** The identity fixes all elements. Since $|X| = 6$,

$$\chi_\pi(e) = 6.$$

- **Class (12):** This permutation swaps $1 \leftrightarrow 2$ and fixes 3. For a pair (i, j) to be fixed, both i and j must be fixed points of (12). The only fixed point is 3. However, elements in X must have distinct indices ($i \neq j$). We cannot form a pair $(3, 3)$ because i must not equal j. Thus, there are no pairs (i, j) with $i \neq j$ formed solely from the fixed point set $\{3\}$.

$$\chi_\pi((12)) = 0.$$

- **Class (123):** This permutation cycles $1 \to 2 \to 3 \to 1$. It has no fixed points in $\{1, 2, 3\}$. Consequently, it cannot fix any pair (i, j).

$$\chi_\pi((123)) = 0.$$

The character is $\chi_\pi = (6, 0, 0)$.

(b) Decomposition into Irreducible Characters

The irreducible characters of S_3 are:

- Trivial $\mathbf{1}$: $(1, 1, 1)$
- Sign ε: $(1, -1, 1)$
- Standard V: $(2, 0, -1)$

We calculate the multiplicities $n_k = \langle \chi_\pi, \chi_k \rangle$:

1. **Multiplicity of $\mathbf{1}$:**

$$n_{\mathbf{1}} = \frac{1}{6}(1 \cdot 6 \cdot 1 + 3 \cdot 0 \cdot 1 + 2 \cdot 0 \cdot 1) = \frac{6}{6} = 1.$$

2. **Multiplicity of ε:**

$$n_\varepsilon = \frac{1}{6}(1 \cdot 6 \cdot 1 + 3 \cdot 0 \cdot (-1) + 2 \cdot 0 \cdot 1) = \frac{6}{6} = 1.$$

3. **Multiplicity of V:**

$$n_V = \frac{1}{6}(1 \cdot 6 \cdot 2 + 3 \cdot 0 \cdot 0 + 2 \cdot 0 \cdot (-1)) = \frac{12}{6} = 2.$$

Check dimensions: $1 \cdot \dim(\mathbf{1}) + 1 \cdot \dim(\varepsilon) + 2 \cdot \dim(V) = 1 + 1 + 2(2) = 6 = \dim(\pi)$. This matches.

Conclusion: The decomposition is:

$$\pi \cong \mathbf{1} \oplus \varepsilon \oplus 2V.$$

Solution to problem 3

(a) Construction of the Subspace W^H

Let $w \in W$. We can write w as a vector $(x_1, x_2, x_3, x_4) \in \mathbb{C}^4$. For w to be in W^H, it must satisfy two conditions:

1. **The Subspace Condition ($w \in W$):** The sum of coordinates must be zero.

$$x_1 + x_2 + x_3 + x_4 = 0 \quad (*).$$

2. **The Invariance Condition (H-fixed):** For all $\sigma \in H$, $\sigma \cdot w = w$. The subgroup $H = S_3$ permutes the indices $\{1, 2, 3\}$ and leaves index 4 fixed. Since H contains all transpositions of $\{1, 2, 3\}$, specifically (12) and (23), we must have:

   - Invariance under (12): $(x_2, x_1, x_3, x_4) = (x_1, x_2, x_3, x_4) \implies x_1 = x_2$.
   - Invariance under (23): $(x_1, x_3, x_2, x_4) = (x_1, x_2, x_3, x_4) \implies x_2 = x_3$.

   Therefore, for w to be invariant under H, the first three coordinates must be equal. Let $x_1 = x_2 = x_3 = \alpha$ for some $\alpha \in \mathbb{C}$.

Now, substitute this into the subspace condition $(*)$:

$$\alpha + \alpha + \alpha + x_4 = 0 \implies 3\alpha + x_4 = 0 \implies x_4 = -3\alpha.$$

Thus, any vector $w \in W^H$ must be of the form:

$$w = (\alpha, \alpha, \alpha, -3\alpha) = \alpha(1, 1, 1, -3).$$

This vector satisfies the sum condition $(1 + 1 + 1 - 3 = 0)$, so it is indeed in W.

The subspace is explicitly:

$$W^H = \mathrm{span}_{\mathbb{C}}\{(1, 1, 1, -3)\}.$$

(b) Dimension and Multiplicity

From the construction in part (a), we see that W^H is spanned by a single non-zero basis vector $v = (1, 1, 1, -3)$. Therefore:

$$\dim(W^H) = 1.$$

The multiplicity of the irreducible representation W in the decomposition of ℓ^H is equal to the dimension of the fixed-point subspace W^H.

Thus, the multiplicity of W in ℓ^H is **1**.